Rajagopal Somaskandan

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Summary

M.Sc. Computer Science (AI/ML Specialization) student and IEEE-published researcher building production-ready Machine Learning, NLP, Computer Vision, and Generative AI systems through internships, research, and hackathons. Hands-on experience with Explainable AI (SHAP), Learning-to-Rank, RAG pipelines, and full-stack ML deployment using Python, PyTorch, and FastAPI. Seeking an entry-level AI/ML Engineer or Machine Learning Intern role.

Education

Sathyabama Institute of Science and Technology Aug 2025 – Jun 2027 (Expected)
M.Sc. Computer Science (AI/ML Specialization) *Chennai, India*

Sathyabama Institute of Science and Technology 2025
B.Sc. Computer Science (AI Specialization) – ***First Class*** *Chennai, India*

Technical Skills

Languages: Python, SQL
Machine Learning: Scikit-learn, XGBoost, LightGBM, PyTorch, TensorFlow, Learning-to-Rank, SHAP
Generative AI: Hugging Face Transformers, LangChain, RAG, Sentence-Transformers (SBERT)
Computer Vision: OpenCV, 3D Gaussian Splatting, COLMAP
Databases: PostgreSQL, MongoDB, FAISS, ChromaDB
Backend & Tools: FastAPI, React, Docker, Git, Linux, MLflow

Experience

Production AI Developer (Intern) Mar 2026 – Jun 2026
InnoXR Labs Pvt. Ltd. *Chennai, India*

- Built an end-to-end **ML pipeline** for single-video **3D reconstruction** using **COLMAP**, **Gaussian Splatting**, and **PyTorch** (MonoSplat).
- Developed **GPU-accelerated preprocessing pipelines** for large-scale video processing, and rewrote the rendering pipeline for training stability.
- Built **FastAPI** backend services and integrated a **Three.js** viewer for real-time 3D visualization; benchmarked on Tanks and Temples.

Data Analyst Intern Jul 2024 – Oct 2024
India Advocacy *Chennai, India*

- Built **Java and Selenium ETL pipelines** processing over **1M records**.
- Performed large-scale **data cleaning and preprocessing** to generate structured analytical datasets.

Projects

SourceUp: Explainable Supplier Ranking System Research Project – Submitted, IEEE ICASET

- Built a **supplier recommendation system** using **SBERT**, **FAISS**, and **LightGBM LambdaRank** over **1.27M supplier records**.
- Created a weak-supervision benchmark with **23K+ supplier–query pairs**, achieving **NDCG@10 of 0.886**; evaluated with **MAP** and **MRR**.
- Integrated **SHAP** explainability and fairness-aware ranking for transparent recommendations.

FinGuard Pro: Explainable Fraud Detection IEEE ICAAIC 2025

- Developed a real-time **fraud detection pipeline** using **XGBoost**, **LightGBM**, and **CatBoost** on **100K transactions**.
- Achieved **94% recall** and **ROC-AUC of 0.9951** with **SHAP**-based explanations.
- Published the work at **IEEE ICAAIC 2025**.

CareerGenie: Multi-Agent Career Platform Hackathon Winner, Industry Innovation '26

- Built a **RAG-powered career platform** using **FastAPI**, **React**, **ChromaDB**, and **LLMs**.
- Developed a **Learning-to-Rank** recommendation engine and multi-agent workflow for personalized career guidance.

- Won **Best Industry Innovation Award** at a national-level hackathon (63+ competing teams).

S.A.F.E: Privacy-Preserving Crowd Anomaly Detection Design Thinking Project, 2026

- Evaluated **five anomaly detection models** across **three benchmark datasets** (ETH/UCY, Zara1/Zara2, Mall), selecting **Median Absolute Deviation (MAD)** for best cross-dataset generalization at **ROC-AUC 0.8391** while cutting false positives by **40–60%**.
- Built a privacy-preserving crowd anomaly detection system with **FastAPI** and real-time explainable alerts.

Publications & Conferences

FinGuard Pro: Explainable Fraud Detection | *IEEE ICAAIC 2025* Published

Achievements & Hackathons

Industry Innovation Hackathon | *Sathyabama Institute of Science and Technology – National Level* Mar 2026

- Won **Best Industry Innovation Award** at Industry Innovation '26 competing against **63+ teams**.

Origin 24-Hour Hackathon | *SIMATS Engineering, Chennai* Apr 2026

- Placed **4th** at Origin 24-Hour Hackathon.

Certifications

AI & ML | *Space Zee Technologies* Jul 2024
Python Core | *Space Zee Technologies* Feb 2024

Additional Experience

Security Trooper – National Service Jan 2020 – Jan 2022
Singapore Armed Forces *Singapore*

- Completed mandatory national service; developed operational discipline and structured data reporting under time-critical conditions, including COVID-19 response support.

Sales Associate (Part-Time) Feb 2022 – Aug 2022
Bata *Singapore*

- Managed customer interactions, billing, and inventory in a high-volume retail environment.